Solution

Radius of the sphere R = 1.5 cm

mass m = 0.038 kg

Volume of the sphere $V=\frac{4}{3}\pi R^3$

$=\frac{4}{3}\times(3.14)\times(1.5\times10^{-2})^3 \;\; = 1.413\times10^{-5}\text{m}^3$

Therefore, density

$$\rho=\frac{m}{V}=\frac{0.038\,\text{kg}}{1.413\times10^{-5}\,\text{m}^3}=2690\ \text{kg m}^{-3}$$

Hence, the specific gravity of the sphere

$$=\frac{2690}{1000}=2.69$$

7.3.2 Pressure due to fluid column at rest

A mountaineer climbing the mountain is able to experience a decrease in pressure with altitude. A person jumping into the swimming pool always realizes an increase in pressure with depth below the water surface. In both the cases, the pressure encountered by the mountaineer and diver is usually due to the hydrostatic pressure, because they are due to fluids that are static.

In order to understand the increase in pressure with depth below the water surface, consider a water sample of cross sectional area A in the form of a cylinder. Let h_1 and h_2 be the depths from the air-water interface to level 1 and level 2 of the cylinder respectively as shown in Figure 7.10(a). Let F_1 be the force acting downwards on level 1 and F_2 be the force acting upwards on level 2 such that $F_1 = P_1 A$ and $F_2 = P_2 A$. Let us assume the mass of the sample to be m and under equilibrium condition, the total upward force (F_2) is balanced by the total downward force (F_1 + mg). In other words, the gravitational force will act downward which is being exactly balanced by the difference between the force $F_2 - F_1$.

$$F_2 - F_1 = mg = F_G \qquad (7.15)$$

Let ρ be the density of the water, then the mass of water available in the sample element is

$$m=\rho V = \rho\, A\, (h_2-h_1)$$
$$V=A\,(h_2-h_1)$$

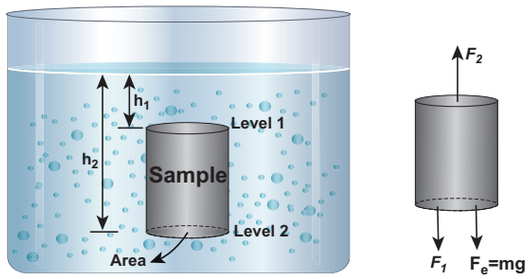


Figure 7.10 (a) A sample of water with base area A in a static fluid with its forces in equilibrium

Hence gravitational force,

$$F_G = \rho\, A\, (h_2 - h_1)\, g$$

On substituting the value of F_G in equation (7.15)

$$F_2=F_1+m\,g$$
$$P_2A = P_1A+\rho A(h_2-h_1)g$$

Cancelling out A on both sides,

$$P_2 = P_1 + \rho(h_2-h_1)g \qquad (7.16)$$

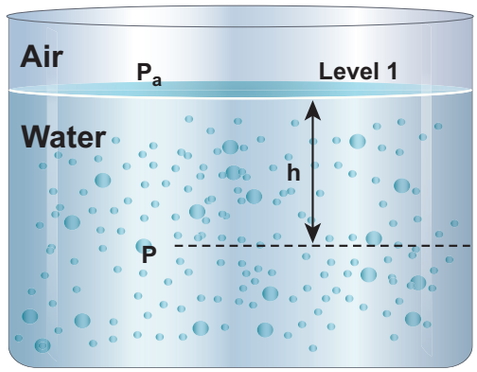


Figure 7.10 (b) The pressure (P) at a depth (h) below the water surface

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